

C.2 CONTACT MAPS: STATISTICAL SIGNIFICANCE TESTS AND NULL MODELS

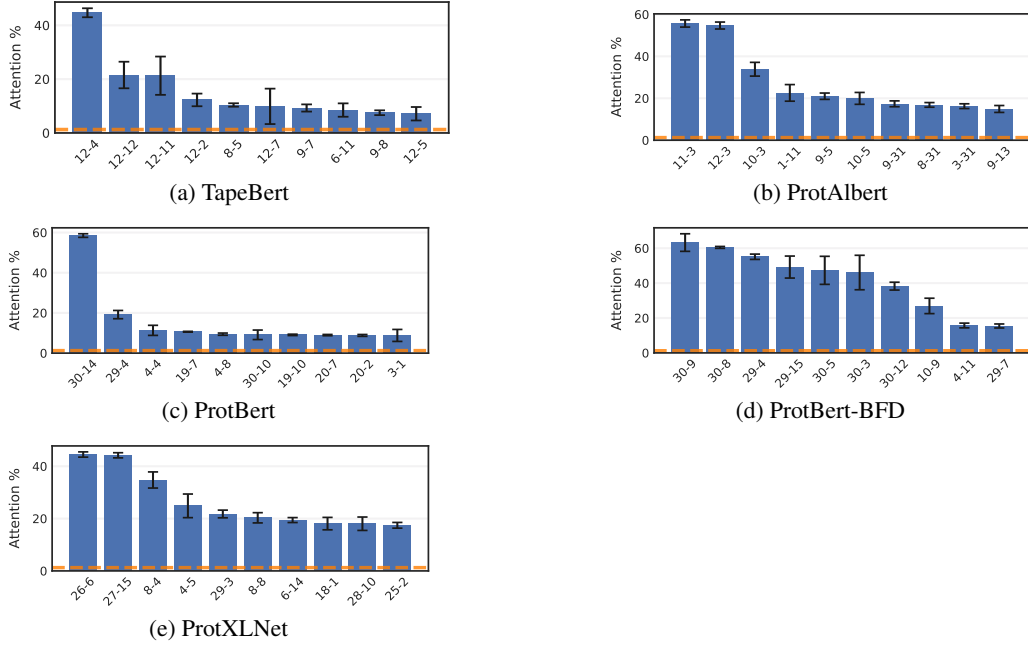


Figure 11: Top 10 heads (denoted by $\langle \text{layer} \rangle - \langle \text{head} \rangle$) for each model based on the proportion of attention aligned with contact maps [95% conf. intervals]. The differences between the attention proportions and the background frequency of contacts (orange dashed line) are statistically significant ($p < 0.00001$). Bonferroni correction applied for both confidence intervals and tests (see App. B.2).

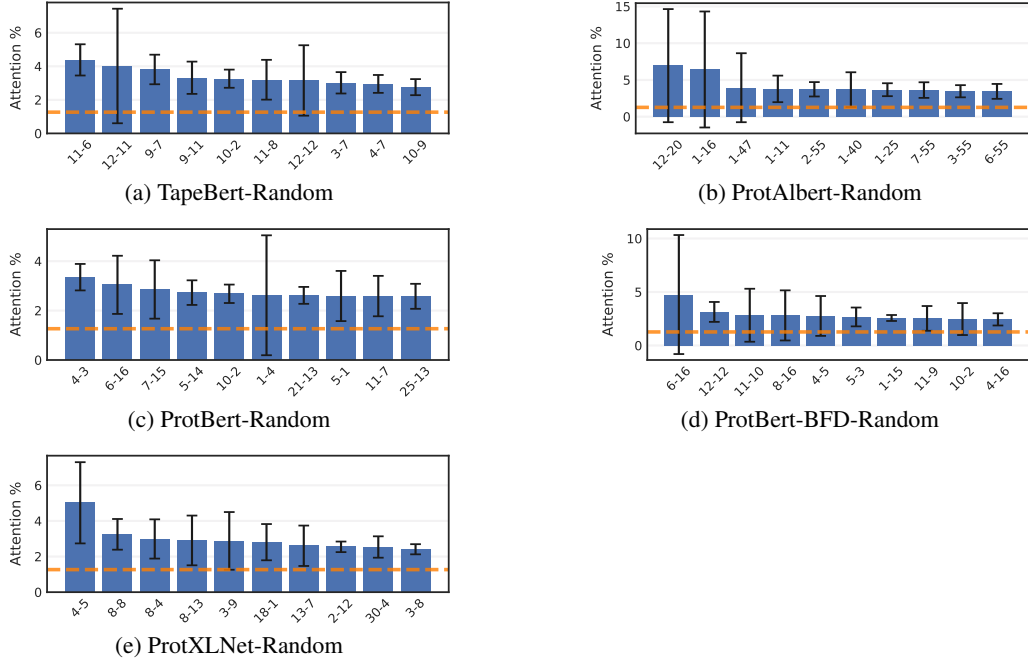


Figure 12: Top-10 contact-aligned heads for **null** models. See Appendix B.2 for details.

C.3 BINDING SITES: STATISTICAL SIGNIFICANCE TESTS AND NULL MODEL

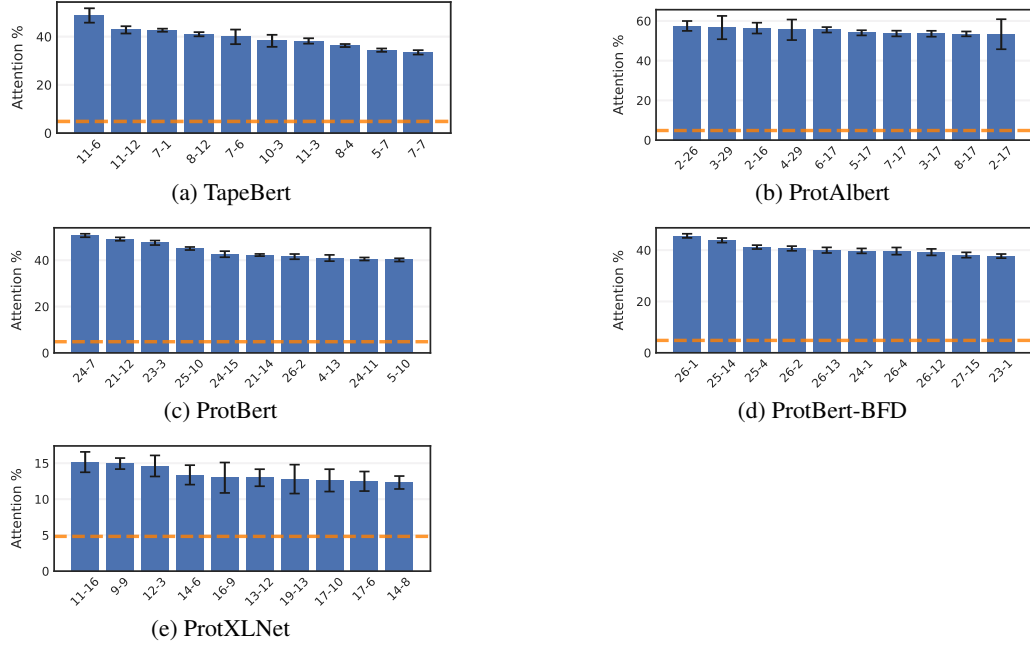


Figure 13: Top 10 heads (denoted by $\langle \text{layer} \rangle - \langle \text{head} \rangle$) for each model based on the proportion of attention focused on binding sites [95% conf. intervals]. Differences between attention proportions and the background frequency of binding sites (orange dashed line) are all statistically significant ($p < 0.00001$). Bonferroni correction applied for both confidence intervals and tests (see App. B.2).

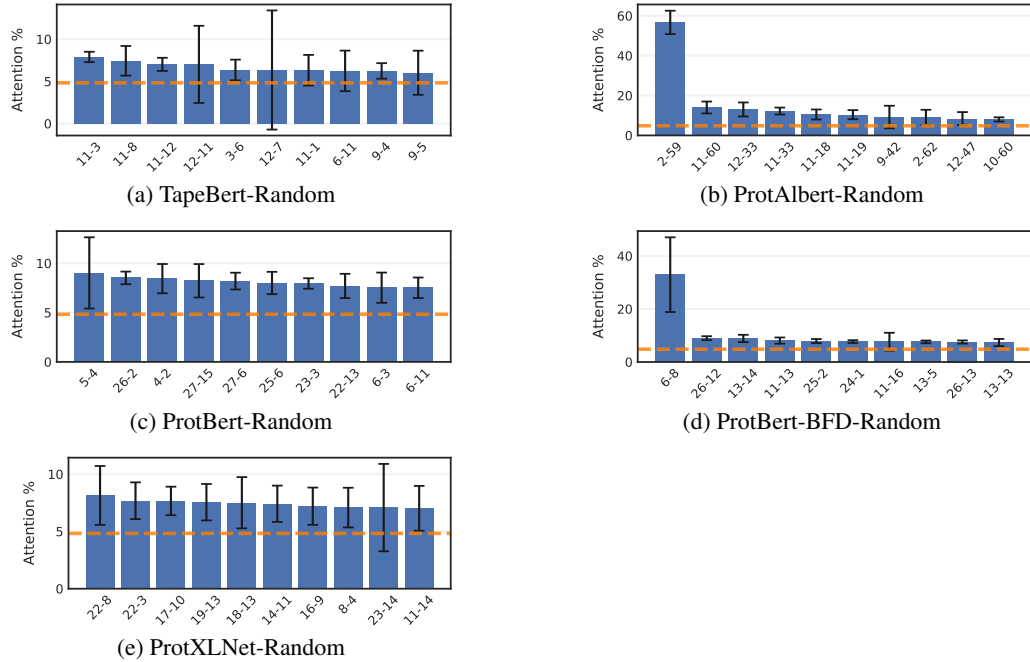


Figure 14: Top-10 heads most focused on binding sites for **null** models. See Appendix B.2 for details.

C.4 POST-TRANSLATIONAL MODIFICATIONS (PTMs)

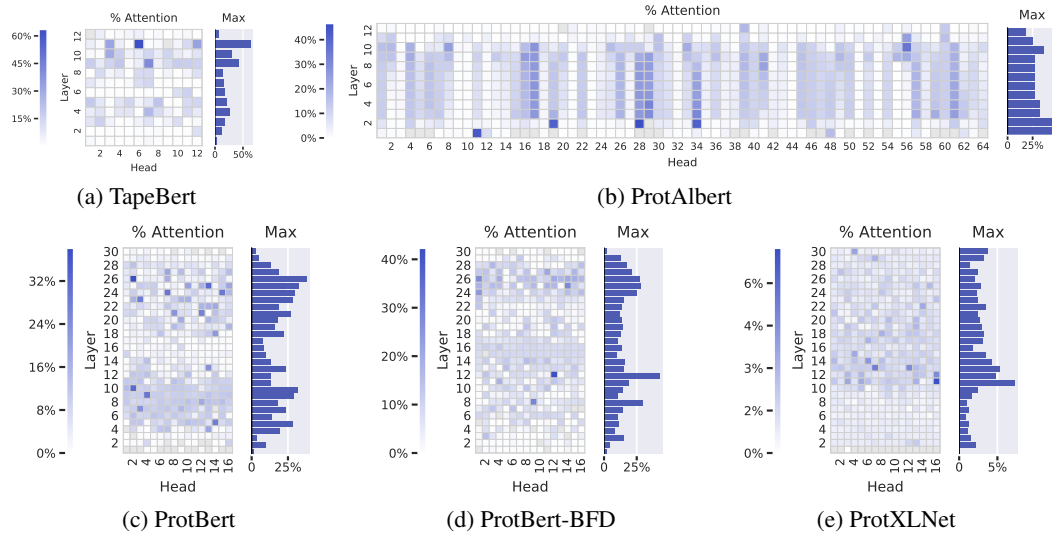


Figure 15: Percentage of each head’s attention that is focused on post-translational modifications.